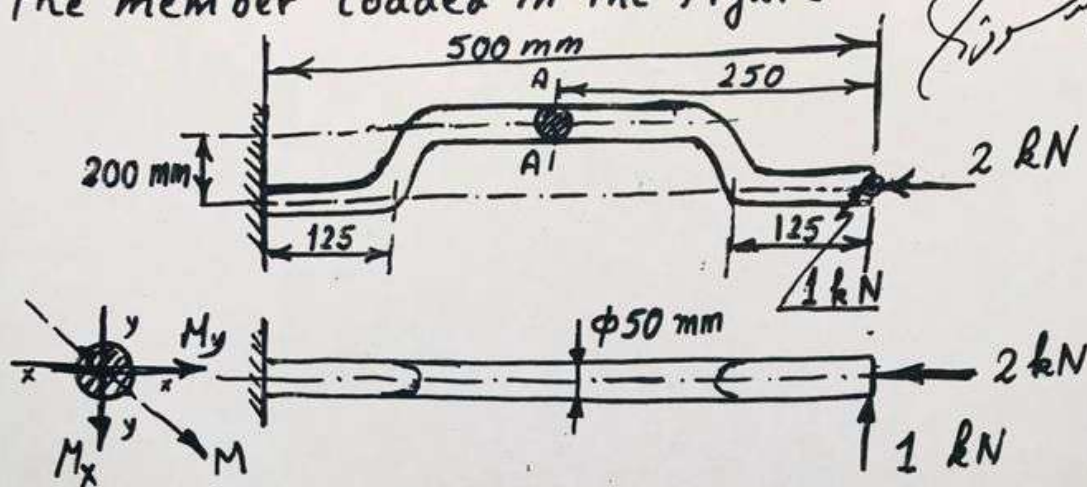


Calculate the maximum normal stress and the maximum shear stress at section A-A in the member loaded in the figure.



$$T = 1 \times 0,2 = 0,2 \text{ kN.m} \leftarrow \text{due to } 1 \text{ kN load.}$$

$$M_x = 2 \times 0,2 = 0,4 \text{ kN.m} \leftarrow \text{due to } 2 \text{ kN load.}$$

$$M_y = 1 \times 0,25 = 0,25 \text{ kN.m} \leftarrow \text{due to } 1 \text{ kN load.}$$

$$M = \sqrt{M_x^2 + M_y^2} = \sqrt{(0,4)^2 + (0,25)^2} = 0,472 \text{ kN.m}$$

$$\sigma = -\frac{F}{A} - \frac{M}{I}y = -\frac{2 \times 10^3}{\frac{\pi}{4}(0,05)^2} - \frac{0,472(0,025) \times 10^3}{\frac{\pi}{64}(0,05)^4}$$

$$= -39,5 \text{ MPa}$$

$$\tau_{xy} = \frac{T}{J}r = \frac{16T}{\pi d^3} = \frac{16(0,2) \times 10^3}{\pi(0,05)^3} = 8,15 \text{ MPa}$$

$$\sigma_{\max/\min} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{\min} = -\frac{39,5}{2} - \sqrt{\left(\frac{39,5}{2}\right)^2 + (8,15)^2} = -41,1 \text{ MPa} \quad \text{[Compression]}$$

$$\tau_{\max} = \sqrt{\left(\frac{39,5}{2}\right)^2 + (8,15)^2} = 21,4 \text{ MPa}$$

$\Rightarrow \sigma_{\min}$ is the maximum numerical normal stress.